

81 - Search In Rotated Sorted Array II

Like previous problem but with duplicates.

① Search in $[0, n - 1]$ $n = \text{len}(A)$

3 4 5 0 1
l m r

3 4 0 1 2
l m r

3 4 5 6 7 0 1 2
l m r $t = 7$

① Determine what sorted array = first or second subarray it belongs to

$\text{nums}[l]$

3 4 5 0 1
l m r

$\text{nums}[l] < \text{nums}[m]$

$\rightarrow \text{nums}[l] <= t < \text{nums}[m]$

$r = m - 1$

else

$l = m + 1$

3 4 0 1 2
l m r

$\text{nums}[m] < \text{nums}[r]$

$\rightarrow \text{nums}[m] < t <= \text{nums}[r]$

$l = m + 1$

else

$r = m - 1$

Dupe Handling

1 0 1 1 1
l m r

- If not the target ship, can't tell what subarray $\text{nums}[m]$ is part of